

Fundamental of AI Unit-2**GTU Syllabus:****Basics of Machine Learning:**

- Learning from examples
- Forms of Learning -Supervised learning, Unsupervised learning, Reinforcement learning
- Simple Models –Linear regression, Logistic regression, Support Vector Machines (SVM) etc.

Machine Learning:

Machine learning is a subset of AI, which enables the machine to automatically learn from data, improve performance from past experiences, and make predictions. Machine learning contains a set of algorithms that work on a huge amount of data. Data is fed to these algorithms to train them, and on the basis of training, they build the model & perform a specific task.

Machine learning real life examples:

Here are some examples of machine learning applications across different domains:

1. Image Recognition

- **Example:** Face recognition in smartphones (e.g., Face ID)
- **Model Type:** Convolutional Neural Networks (CNNs)
- **Use Case:** Security, tagging people in photos (Facebook, Google Photos)

2. Natural Language Processing (NLP)

- **Example:** Chatbots and virtual assistants (e.g., ChatGPT, Siri, Alexa)
- **Model Type:** Transformer models (GPT, BERT)
- **Use Case:** Customer support, content generation

3. Fraud Detection

- **Example:** Credit card fraud detection
- **Model Type:** Anomaly detection, Decision Trees, Neural Networks
- **Use Case:** Banking and finance to identify unusual transactions

4. Recommendation Systems

- **Example:** Netflix recommending movies based on watch history
- **Model Type:** Collaborative filtering, Matrix Factorization
- **Use Case:** E-commerce (Amazon), streaming services (Spotify, YouTube)

5. Autonomous Vehicles

- **Example:** Self-driving cars (Tesla, Waymo)
- **Model Type:** Reinforcement Learning, Deep Learning
- **Use Case:** Navigation, object detection, decision-making

6. Medical Diagnosis

- **Example:** Detecting tumors in X-rays using AI
- **Model Type:** Deep Neural Networks (DNNs), CNNs
- **Use Case:** Early disease detection, AI-assisted healthcare

7. Spam Filtering

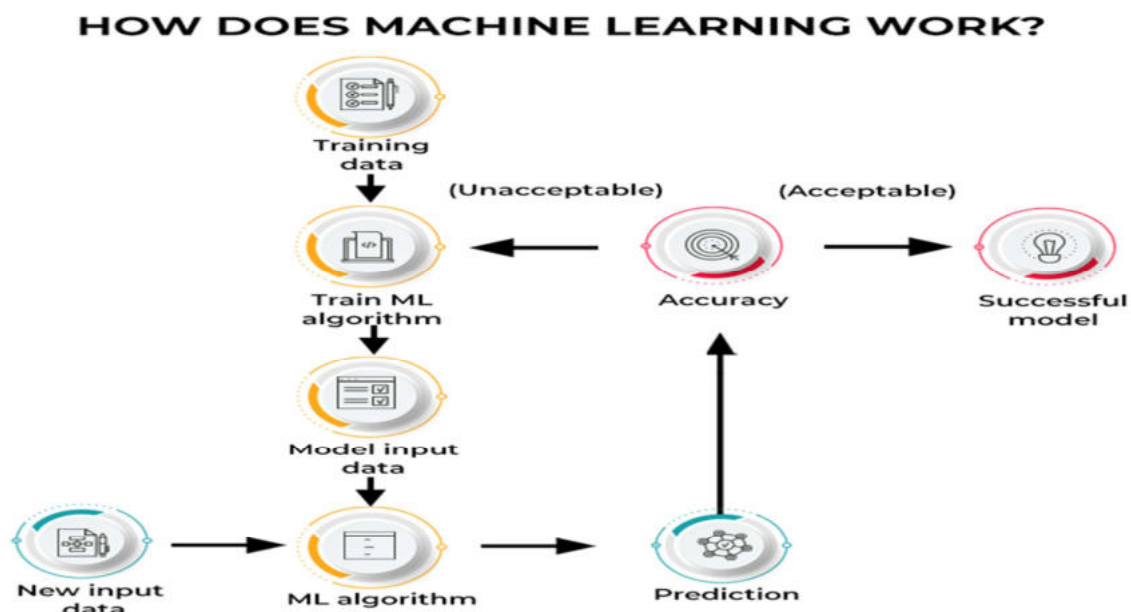
- **Example:** Gmail automatically classifying spam emails
- **Model Type:** Naïve Bayes, Logistic Regression
- **Use Case:** Email security and filtering unwanted messages

8. Stock Market Prediction

- **Example:** AI models predicting stock trends
- **Model Type:** LSTMs, Random Forest, Reinforcement Learning
- **Use Case:** Investment and trading strategies

How does machine learning work?

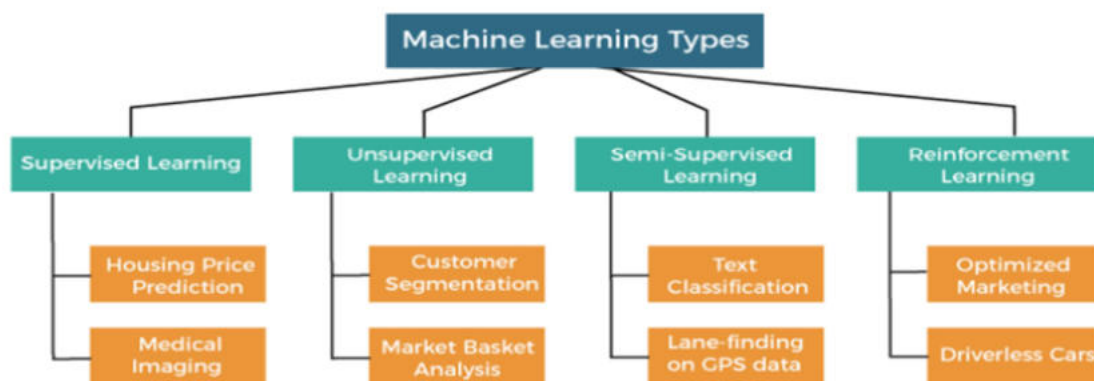
Machine learning algorithms are molded on a training dataset to create a model. As new input data is introduced to the trained ML algorithm, it uses the developed model to make a prediction.



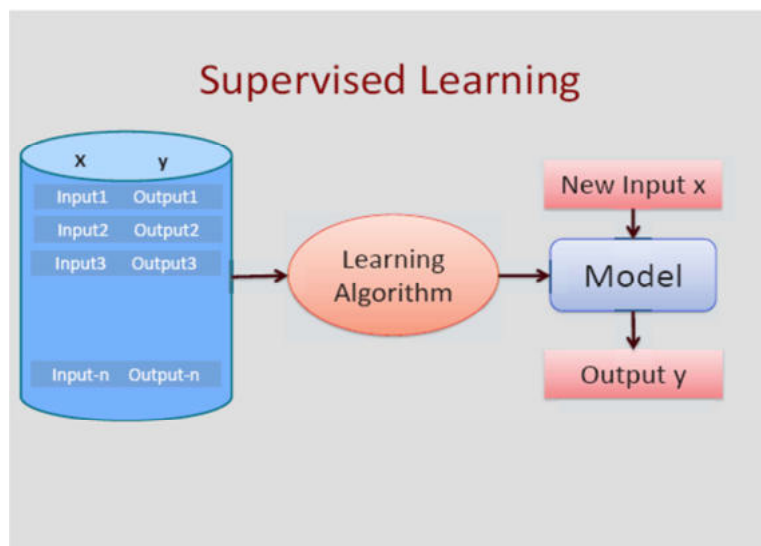
Machine Learning Types:

Based on the methods and way of learning, machine learning is divided into mainly four types, which are:

1. Supervised Machine Learning
2. Unsupervised Machine Learning
3. Semi-Supervised Machine Learning
4. Reinforcement Learning



1. Supervised Machine Learning



In supervised learning, we have data and data comprises of input and the corresponding output. For every data instance we can have the input x and the corresponding output y . And, from this the machine learning system will build a model so that given a new observation x will try to find out what is the corresponding y .

This is called supervised learning because for every instance we tell what is the output. So, this is called labeled data.

In this type of learning, we are given:

- a set of input features
- a target feature (y)
- a set of training examples where the values for the input features and the target features are given for each example
- a new example, where only the values for the input features are given

And we have to Predict the values for the target features for the new example.

- classification when Y is discrete
- regression when Y is continuous

As its name suggests, Supervised machine learning is based on supervision. It means in the supervised learning technique, we train the machines using the "labelled" dataset, and based on the training, the machine predicts the output. Here, the labelled data specifies that some of the inputs are already mapped to the output. More precisely, we can say; first, we train the machine with the input and corresponding output, and then we ask the machine to predict the output using the test dataset.

Let's understand supervised learning with an example. Suppose we have an input dataset of cats and dog images. So, first, we will provide the training to the machine to understand the images, such as the **shape & size of the tail of cat and dog, Shape of eyes, colour, height (dogs are taller, cats are smaller), etc.** After completion of training, we input the picture of a cat and ask the machine to identify the object and predict the output. Now, the machine is well trained, so it will check all the features of the object, such as height, shape, colour, eyes, ears, tail, etc., and find that it's a cat. So, it will put it in the Cat category. This is the process of how the machine identifies the objects in Supervised Learning.

The main goal of the supervised learning technique is to map the input variable(x) with the output variable(y). Some real-world applications of supervised learning are **Risk Assessment, Fraud Detection, Spam filtering, etc.**

Categories of Supervised Machine Learning

Supervised machine learning can be classified into two types of problems, which are given below:

- **Classification**
- **Regression**

a) Classification

Classification algorithms are used to solve the classification problems in which the output variable is categorical, such as **"Yes" or No, Male or Female, Red or Blue, etc.** The

classification algorithms predict the categories present in the dataset. Some real-world examples of classification algorithms are **Spam Detection, Email filtering, etc.**

Some popular classification algorithms are given below:

- **Random Forest Algorithm**
- **Decision Tree Algorithm**
- **Logistic Regression Algorithm**
- **Support Vector Machine Algorithm**

b) Regression

Regression algorithms are used to solve regression problems in which there is a linear relationship between input and output variables. These are used to predict continuous output variables, such as market trends, weather prediction, etc.

Some popular Regression algorithms are given below:

- **Simple Linear Regression Algorithm**
- **Multivariate Regression Algorithm**
- **Decision Tree Algorithm**
- **Lasso Regression**

Advantages and Disadvantages of Supervised Learning

Advantages:

- Since supervised learning work with the labelled dataset so we can have an exact idea about the classes of objects.
- These algorithms are helpful in predicting the output on the basis of prior experience.

Disadvantages:

- These algorithms are not able to solve complex tasks.
- It may predict the wrong output if the test data is different from the training data.
- It requires lots of computational time to train the algorithm.

Applications of Supervised Learning

Some common applications of Supervised Learning are given below:

- **Image Segmentation:**

Supervised Learning algorithms are used in image segmentation. In this process, image classification is performed on different image data with pre-defined labels.

- **Medical**

Diagnosis:

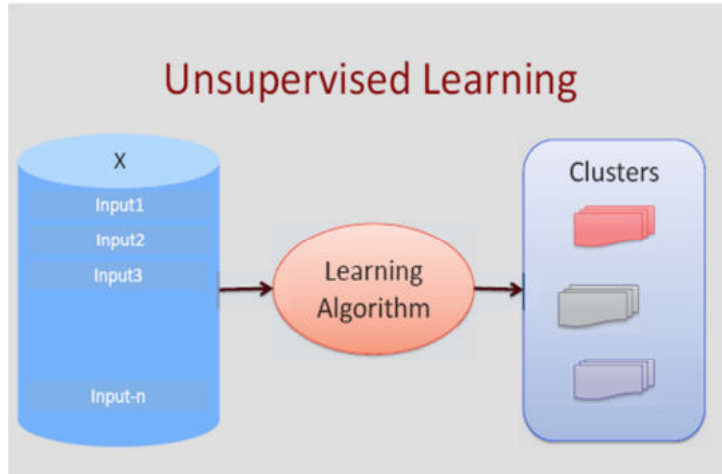
Supervised algorithms are also used in the medical field for diagnosis purposes. It is done by using medical images and past labelled data with labels for disease conditions. With such a process, the machine can identify a disease for the new patients.

- **Fraud Detection** - Supervised Learning classification algorithms are used for identifying fraud transactions, fraud customers, etc. It is done by using historic data to identify the patterns that can lead to possible fraud.

- **Spam detection** - In spam detection & filtering, classification algorithms are used. These algorithms classify an email as spam or not spam. The spam emails are sent to the spam folder.

- **Speech Recognition** - Supervised learning algorithms are also used in speech recognition. The algorithm is trained with voice data, and various identifications can be done using the same, such as voice-activated passwords, voice commands, etc.

2. Unsupervised Machine Learning



- In unsupervised learning you are only given x , there is no label to the data. And given the different data points you may like to for example cluster them or summarize them or find some patterns in them.
- In this, we only have x 's. We have different x 's; x_1, x_2, x_n this is the data. And the learning algorithm will produce clusters or group this data.
- Based on the similarity of the data items to each other we can find out certain groups among the data. So that is called Unsupervised learning.

Unsupervised learning is different from the Supervised learning technique; as its name suggests, there is no need for supervision. It means, in unsupervised machine learning, the

machine is trained using the unlabeled dataset, and the machine predicts the output without any supervision.

In unsupervised learning, the models are trained with the data that is neither classified nor labelled, and the model acts on that data without any supervision.

The main aim of the unsupervised learning algorithm is to group or categories the unsorted dataset according to the similarities, patterns, and differences. Machines are instructed to find the hidden patterns from the input dataset.

Let's take an example to understand it more precisely; suppose there is a basket of fruit images, and we input it into the machine learning model. The images are totally unknown to the model, and the task of the machine is to find the patterns and categories of the objects.

So, now the machine will discover its patterns and differences, such as colour difference, shape difference, and predict the output when it is tested with the test dataset.

Categories of Unsupervised Machine Learning

Unsupervised Learning can be further classified into two types, which are given below:

- **Clustering**
- **Association**

1) Clustering

The clustering technique is used when we want to find the inherent groups from the data. It is a way to group the objects into a cluster such that the objects with the most similarities remain in one group and have fewer or no similarities with the objects of other groups. An example of the clustering algorithm is grouping the customers by their purchasing behaviour.

Some of the popular clustering algorithms are given below:

- **K-Means Clustering algorithm**
- **Mean-shift algorithm**
- **DBSCAN Algorithm**
- **Principal Component Analysis**
- **Independent Component Analysis**

2) Association

Association rule learning is an unsupervised learning technique, which finds interesting relations among variables within a large dataset. The main aim of this learning algorithm is to find the dependency of one data item on another data item and map those variables

accordingly so that it can generate maximum profit. This algorithm is mainly applied in **Market Basket analysis, Web usage mining, continuous production**, etc.

Some popular algorithms of Association rule learning are **Apriori Algorithm, Eclat, FP-growth algorithm**.

Advantages and Disadvantages of Unsupervised Learning Algorithm

Advantages:

- These algorithms can be used for complicated tasks compared to the supervised ones because these algorithms work on the unlabeled dataset.
- Unsupervised algorithms are preferable for various tasks as getting the unlabeled dataset is easier as compared to the labelled dataset.

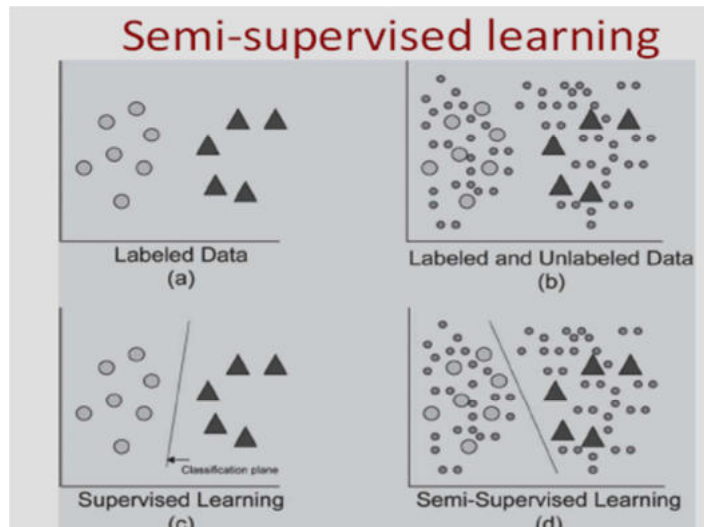
Disadvantages:

- The output of an unsupervised algorithm can be less accurate as the dataset is not labelled, and algorithms are not trained with the exact output in prior.
- Working with Unsupervised learning is more difficult as it works with the unlabelled dataset that does not map with the output.

Applications of Unsupervised Learning

- **Network Analysis:** Unsupervised learning is used for identifying plagiarism and copyright in document network analysis of text data for scholarly articles.
- **Recommendation Systems:** Recommendation systems widely use unsupervised learning techniques for building recommendation applications for different web applications and e-commerce websites.
- **Anomaly Detection:** Anomaly detection is a popular application of unsupervised learning, which can identify unusual data points within the dataset. It is used to discover fraudulent transactions.
- **Singular Value Decomposition:** Singular Value Decomposition or SVD is used to extract particular information from the database. For example, extracting information of each user located at a particular location.

3. Semi-Supervised Learning



Semi-Supervised learning is a type of Machine Learning algorithm that lies between Supervised and Unsupervised machine learning. It represents the intermediate ground between Supervised (With Labelled training data) and Unsupervised learning (with no labelled training data) algorithms and uses the combination of labelled and unlabeled datasets during the training period.

Although Semi-supervised learning is the middle ground between supervised and unsupervised learning and operates on the data that consists of a few labels, it mostly consists of unlabeled data. As labels are costly, but for corporate purposes, they may have few labels. It is completely different from supervised and unsupervised learning as they are based on the presence & absence of labels.

To overcome the drawbacks of supervised learning and unsupervised learning algorithms, the concept of Semi-supervised learning is introduced. The main aim of [semi-supervised learning](#) is to effectively use all the available data, rather than only labelled data like in supervised learning. Initially, similar data is clustered along with an unsupervised learning algorithm, and further, it helps to label the unlabeled data into labelled data. It is because labelled data is a comparatively more expensive acquisition than unlabeled data.

We can imagine these algorithms with an example. Supervised learning is where a student is under the supervision of an instructor at home and college. Further, if that student is self-analysing the same concept without any help from the instructor, it comes under unsupervised learning. Under semi-supervised learning, the student has to revise himself after analyzing the same concept under the guidance of an instructor at college.

Advantages and disadvantages of Semi-supervised Learning

Advantages:

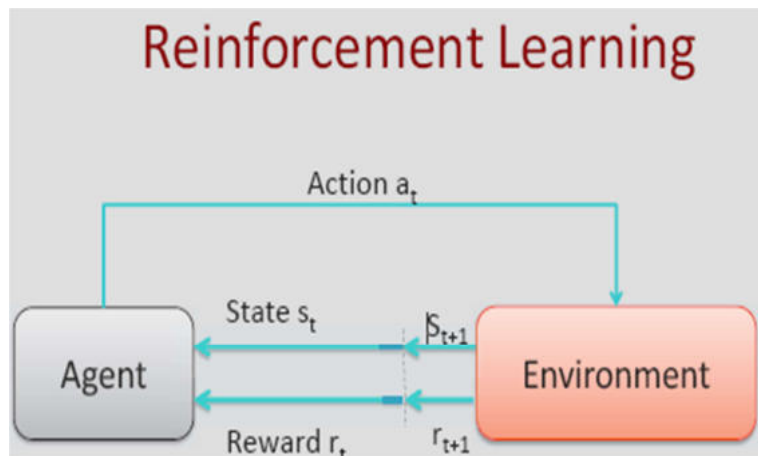
- It is simple and easy to understand the algorithm.
- It is highly efficient.

- It is used to solve drawbacks of Supervised and Unsupervised Learning algorithms.

Disadvantages:

- Iterations results may not be stable.
- We cannot apply these algorithms to network-level data.
- Accuracy is low.

4. Reinforcement Learning



- In reinforcement learning we have an agent which acts in the environment.
- The agent can take action and this action can impact the environment.
- In a particular stage, the agent takes an action and the environment goes to a new state and gives some reward to the agent, that reward may be a positive reward can be a negative reward or penalty or can be nothing at that particular time step.
- But the agent is continually acting in this world.
- And the reinforcement learner will learn a policy. That is, given a state what action to take so that not only the short term reward is optimized, but the overall utility of the agent over its entire time horizon is optimized.

Reinforcement learning works on a feedback-based process, in which an AI agent (A software component) automatically explore its surrounding by hitting & trail, taking action, learning from experiences, and improving its performance. Agent gets rewarded for each good action and get punished for each bad action; hence the goal of reinforcement learning agent is to maximize the rewards.

In reinforcement learning, there is no labelled data like supervised learning, and agents learn from their experiences only.

The [reinforcement learning](#) process is similar to a human being; for example, a child learns various things by experiences in his day-to-day life. An example of reinforcement learning is to play a game, where the Game is the environment, moves of an agent at each step define

states, and the goal of the agent is to get a high score. Agent receives feedback in terms of punishment and rewards.

Due to its way of working, reinforcement learning is employed in different fields such as **Game theory, Operation Research, Information theory, multi-agent systems.**

A reinforcement learning problem can be formalized using **Markov Decision Process(MDP)**. In MDP, the agent constantly interacts with the environment and performs actions; at each action, the environment responds and generates a new state.

Categories of Reinforcement Learning

Reinforcement learning is categorized mainly into two types of methods/algorithms:

- **Positive Reinforcement Learning:** Positive reinforcement learning specifies increasing the tendency that the required behaviour would occur again by adding something. It enhances the strength of the behaviour of the agent and positively impacts it.
- **Negative Reinforcement Learning:** Negative reinforcement learning works exactly opposite to the positive RL. It increases the tendency that the specific behaviour would occur again by avoiding the negative condition.

Real-world Use cases of Reinforcement Learning

- **Video-Games:**
RL algorithms are much popular in gaming applications. It is used to gain super-human performance. Some popular games that use RL algorithms are **AlphaGO** and **AlphaGO Zero**.
- **Resource-Management:**
The "Resource Management with Deep Reinforcement Learning" paper showed that how to use RL in computer to automatically learn and schedule resources to wait for different jobs in order to minimize average job slowdown.
- **Robotics:**
RL is widely being used in Robotics applications. Robots are used in the industrial and manufacturing area, and these robots are made more powerful with reinforcement learning. There are different industries that have their vision of building intelligent robots using AI and Machine learning technology.
- **Text-Mining**
Text-mining, one of the great applications of NLP, is now being implemented with the help of Reinforcement Learning by Salesforce company.

Advantages and Disadvantages of Reinforcement Learning

Advantages

- It helps in solving complex real-world problems which are difficult to be solved by general techniques.
- The learning model of RL is similar to the learning of human beings; hence most accurate results can be found.
- Helps in achieving long term results.

Disadvantage

- RL algorithms are not preferred for simple problems.
- RL algorithms require huge data and computations.
- Too much reinforcement learning can lead to an overload of states which can weaken the results.

Simple Models:*What is Linear Regression?*

Linear regression is a statistical method used to model the relationship between a dependent variable (target) and one or more independent variables (predictors). It is widely used for predictive modeling, trend analysis, and forecasting.

The mathematical equation for simple linear regression (with one independent variable) is:

Formula:

$$Y = b_0 + b_1X + \epsilon$$

Where:

- Y = Dependent variable (target)
- X = Independent variable (feature)
- b_0 = Intercept (bias)
- b_1 = Slope (coefficient)
- ϵ = Error term

If there are multiple independent variables, it becomes **Multiple Linear Regression**:

Formula:

$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n + \epsilon$$

Used when multiple factors influence the dependent variable.

Example: Predicting House Prices using Linear Regression

Let's consider a dataset where we predict house prices based on their size (square footage).

We have historical data on house sizes (in square feet) and their prices (in \$1000s).

House Size (sq ft)	Price (\$1000s)
850	120
900	135
1200	180
1500	225
1750	260
2000	300



Here is the plotted **Linear Regression** graph:

- **Blue dots** represent the actual house price data.
- The **red line** is the best-fit regression line.
- **Green crosses** are predicted prices for new house sizes (800, 1000, 1600, and 2100 sq ft).

What is Logistic Regression?

Logistic regression is a classification algorithm used to predict **binary outcomes** (e.g., Yes/No, 0/1, Pass/Fail). Unlike linear regression, which predicts continuous values, logistic regression predicts probabilities using the **sigmoid function**:

$$P(Y = 1) = \frac{1}{1 + e^{-(b_0 + b_1 X_1 + b_2 X_2 + \dots + b_n X_n)}}$$

- $P(Y = 1)$ is the probability of the outcome being 1.
- The sigmoid function ensures that the output is between 0 and 1.

Types of Logistic Regression

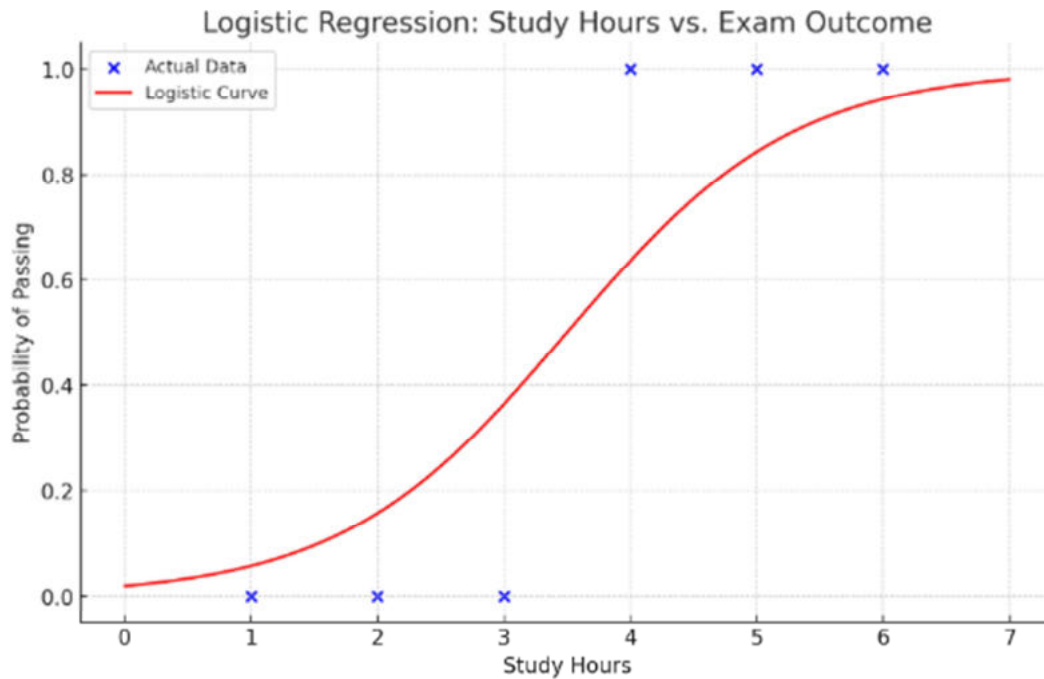
On the basis of the categories, Logistic Regression can be classified into three types:

1. **Binomial**: In binomial Logistic regression, there can be only two possible types of the dependent variables, such as 0 or 1, Pass or Fail, etc.
2. **Multinomial**: In multinomial Logistic regression, there can be 3 or more possible unordered types of the dependent variable, such as “cat”, “dogs”, or “sheep”
3. **Ordinal**: In ordinal Logistic regression, there can be 3 or more possible ordered types of dependent variables, such as “low”, “Medium”, or “High”.

Example: Predicting Whether a Student Passes an Exam

We want to predict if a student **Passes (1) or Fails (0)** based on the number of study hours.

Study Hours	Pass (1) / Fail (0)
1	0
2	0
3	0
4	1
5	1
6	1



Here is the **Logistic Regression** plot:

- **Blue dots** represent actual student results (Pass = 1, Fail = 0).
- The **red sigmoid curve** represents the probability of passing as study hours increase.
- The **decision boundary** (around 3.5 study hours) indicates when the model predicts a student is more likely to pass.

Metrics for Linear Regression

Linear regression aims to model the relationship between a dependent variable y and one or more independent variables X . The key metrics to evaluate the performance of a linear regression model include:

1. Mean Absolute Error (MAE):

- Measures the average magnitude of errors in predictions without considering their direction.
- Formula:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- Lower MAE indicates better model performance.

2. Mean Squared Error (MSE):

- Measures the average of the squared differences between predicted and actual values.
- Formula:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- MSE penalizes larger errors more than MAE.

3. Root Mean Squared Error (RMSE):

- The square root of MSE, providing error in the same unit as the target variable.
- Formula:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- RMSE is more sensitive to outliers than MAE.

4. R-Squared (R^2):

- Measures the proportion of the variance in the dependent variable that is predictable from the independent variable(s).
- Formula:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

- R-squared ranges from 0 to 1, with 1 indicating perfect fit.

5. Adjusted R-Squared:

- Adjusts the R^2 value for the number of predictors, preventing overfitting.
- Formula:

$$\text{Adjusted } R^2 = 1 - \left(\frac{(1 - R^2) \times (n - 1)}{n - p - 1} \right)$$

- p: Number of predictors
- n: Sample size

Principle of Least Squares

- The **Least Squares** method is used to find the best-fitting line (or model) to a set of data points by minimizing the sum of the squares of the differences (errors) between the observed values and the values predicted by the model.
- In simple linear regression, we try to fit a line of the form:

$$\hat{y} = b_0 + b_1x$$

Where:

- $\hat{y}$ is the predicted value,
- x is the independent variable,
- b_0 is the intercept,
- b_1 is the slope.

The "least squares" method finds the values of b_0 and b_1 that minimize the sum of squared errors (SSE).

Equations

1. Least Squares Estimators

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$b_0 = \bar{y} - b_1\bar{x}$$

Where:

- $\bar{x}$ is the mean of x_i ,
- $\bar{y}$ is the mean of y_i .

2. Fitted Regression Line

$$\hat{y}_i = b_0 + b_1x_i$$

3. Residual (Error)

$$e_i = y_i - \hat{y}_i$$

It's the difference between the observed and predicted values.

4. Sum of Squared Errors (SSE)

$$SSE = \sum (y_i - \hat{y}_i)^2 = \sum e_i^2$$

This is what the least squares method minimizes.

Difference Between Linear Regression and Logistic Regression

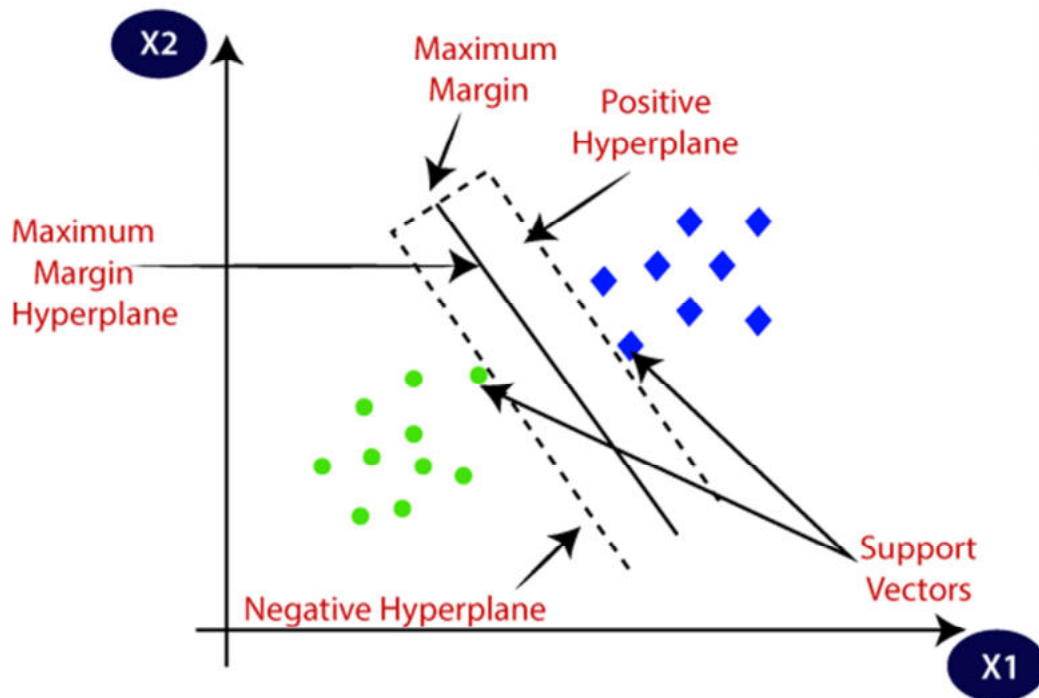
Feature	Linear Regression	Logistic Regression
Type of Problem	Regression (Continuous Output)	Classification (Binary/Categorical Output)
Output Values	Any real number (e.g., house price = \$250,000)	Probability between 0 and 1 (e.g., chance of passing = 0.85)
Equation Used	Formula: $Y = b_0 + b_1X + \epsilon$ Where: • Y = Dependent variable (target) • X = Independent variable (feature) • b_0 = Intercept (bias) • b_1 = Slope (coefficient) • ϵ = Error term	$P(Y=1) = \frac{1}{1 + e^{-(b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n)}}$ • $P(Y=1)$ is the probability of the outcome being 1. • The sigmoid function ensures that the output is between 0 and 1.
Algorithm Used	Ordinary Least Squares (OLS)	Maximum Likelihood Estimation (MLE)
Best Fit Line	Straight line (linear relationship)	S-curve (sigmoid function)
Use Cases	Predicting house prices, stock prices, sales revenue, etc.	Spam detection, disease prediction, credit approval, etc.

Support Vector Machine Algorithm:

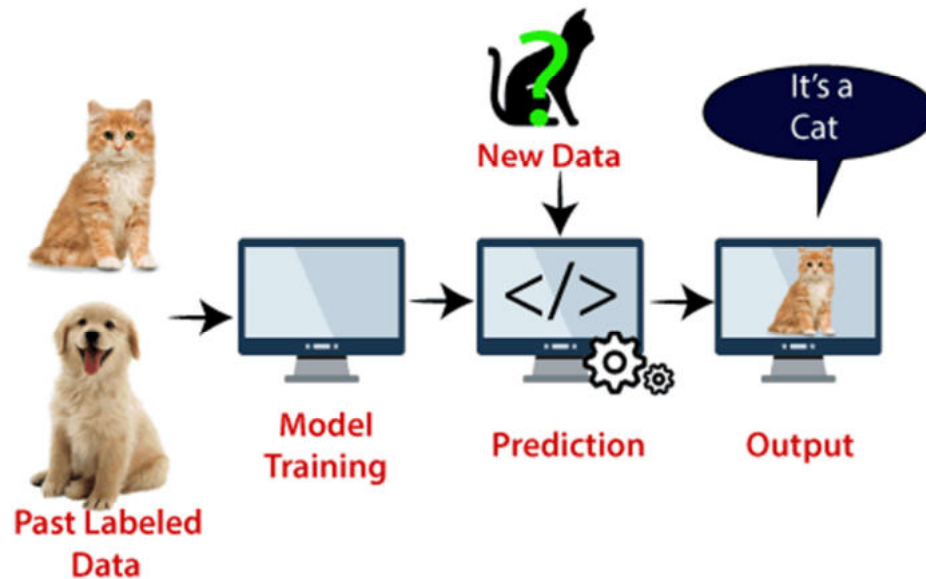
Support Vector Machine or SVM is one of the most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems. However, primarily, it is used for Classification problems in Machine Learning.

The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future. This best decision boundary is called a hyperplane.

SVM chooses the extreme points/vectors that help in creating the hyperplane. These extreme cases are called as support vectors, and hence algorithm is termed as Support Vector Machine. Consider the below diagram in which there are two different categories that are classified using a decision boundary or hyperplane:



Example: SVM can be understood with the example that we have used in the KNN classifier. Suppose we see a strange cat that also has some features of dogs, so if we want a model that can accurately identify whether it is a cat or dog, so such a model can be created by using the SVM algorithm. We will first train our model with lots of images of cats and dogs so that it can learn about different features of cats and dogs, and then we test it with this strange creature. So as support vector creates a decision boundary between these two data (cat and dog) and choose extreme cases (support vectors), it will see the extreme case of cat and dog. On the basis of the support vectors, it will classify it as a cat. Consider the below diagram:



SVM algorithm can be used for **Face detection, image classification, text categorization**, etc.

Types of SVM

SVM can be of two types:

- **Linear SVM:** Linear SVM is used for linearly separable data, which means if a dataset can be classified into two classes by using a single straight line, then such data is termed as linearly separable data, and classifier is used called as Linear SVM classifier.
- **Non-linear SVM:** Non-Linear SVM is used for non-linearly separated data, which means if a dataset cannot be classified by using a straight line, then such data is termed as non-linear data and classifier used is called as Non-linear SVM classifier.

Hyperplane and Support Vectors in the SVM algorithm:

Hyperplane: There can be multiple lines/decision boundaries to segregate the classes in n-dimensional space, but we need to find out the best decision boundary that helps to classify the data points. This best boundary is known as the hyperplane of SVM.

The dimensions of the hyperplane depend on the features present in the dataset, which means if there are 2 features (as shown in image), then hyperplane will be a straight line. And if there are 3 features, then hyperplane will be a 2-dimension plane.

We always create a hyperplane that has a maximum margin, which means the maximum distance between the data points.

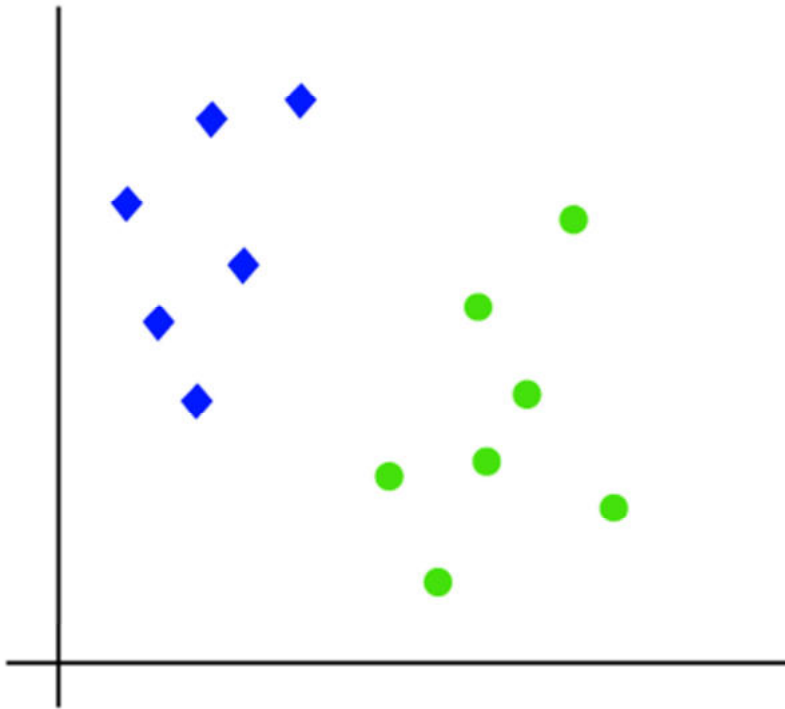
Support Vectors:

The data points or vectors that are the closest to the hyperplane and which affect the position of the hyperplane are termed as Support Vector. Since these vectors support the hyperplane, hence called a Support vector.

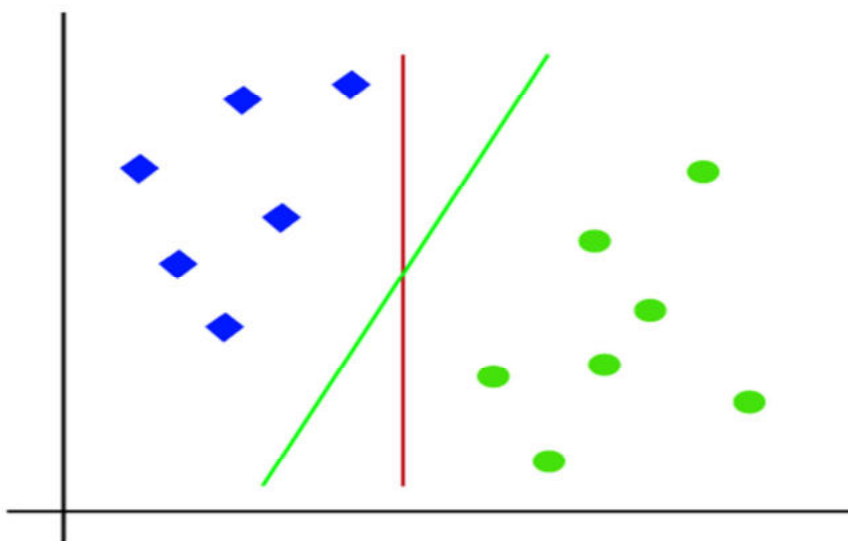
How does SVM works?

Linear SVM:

The working of the SVM algorithm can be understood by using an example. Suppose we have a dataset that has two tags (green and blue), and the dataset has two features x_1 and x_2 . We want a classifier that can classify the pair(x_1 , x_2) of coordinates in either green or blue. Consider the below image:

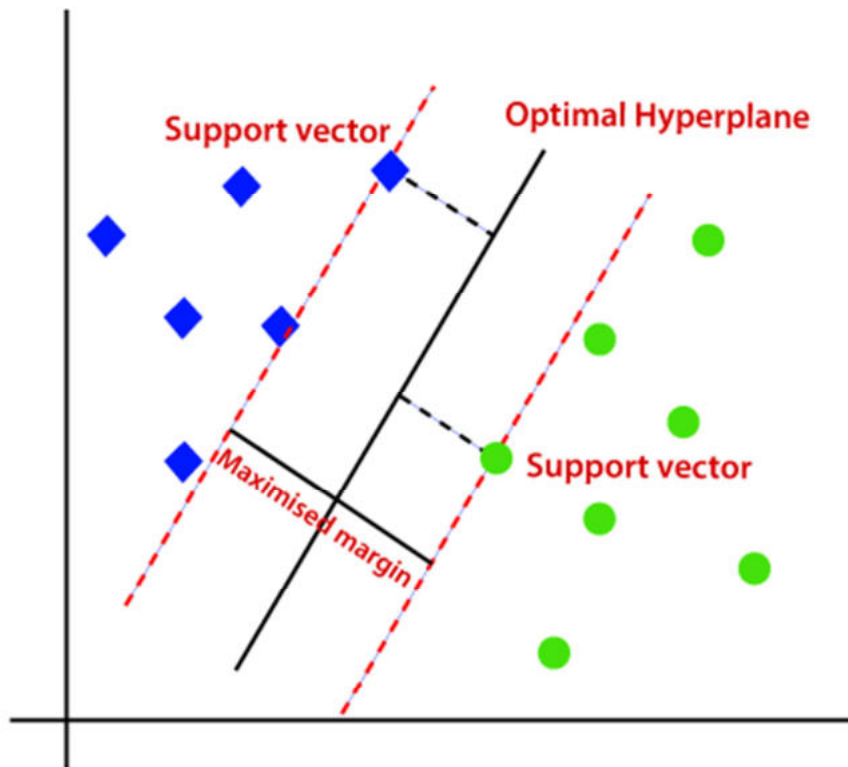


So as it is 2-d space so by just using a straight line, we can easily separate these two classes. But there can be multiple lines that can separate these classes. Consider the below image:



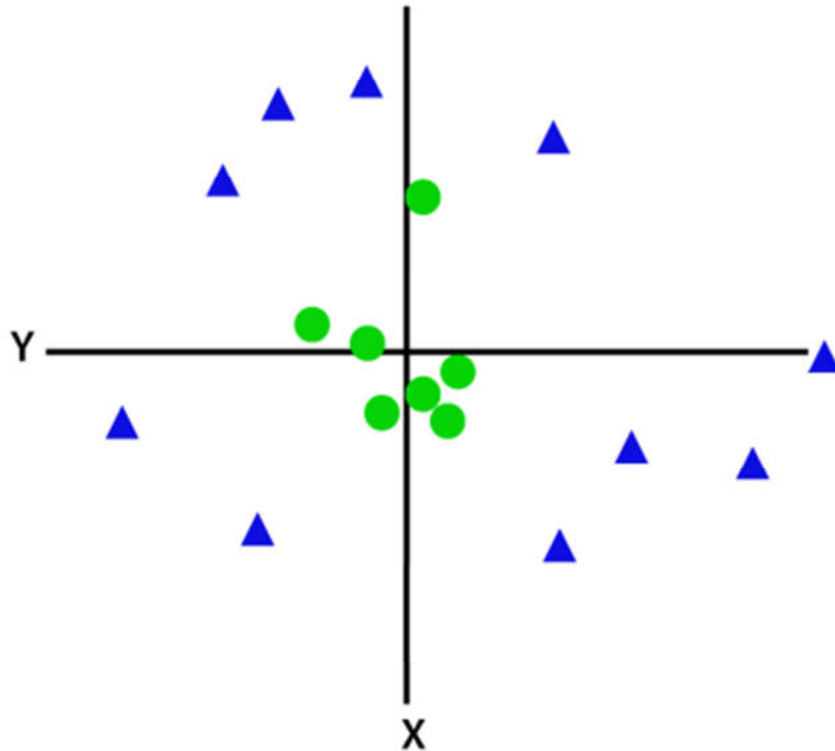
Hence, the SVM algorithm helps to find the best line or decision boundary; this best boundary or region is called as a **hyperplane**. SVM algorithm finds the closest point of the

lines from both the classes. These points are called support vectors. The distance between the vectors and the hyperplane is called as **margin**. And the goal of SVM is to maximize this margin. The **hyperplane** with maximum margin is called the **optimal hyperplane**.



Non-Linear SVM:

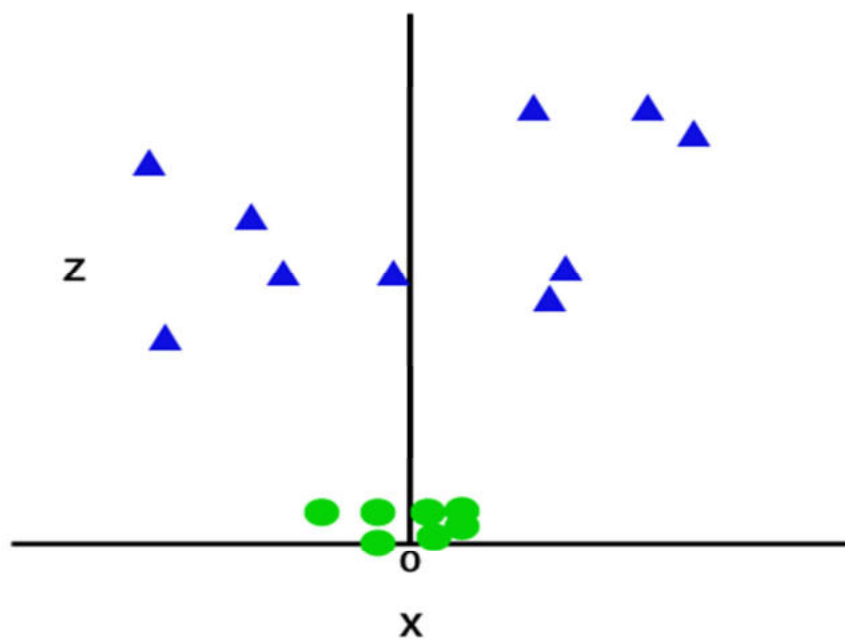
If data is linearly arranged, then we can separate it by using a straight line, but for non-linear data, we cannot draw a single straight line. Consider the below image:



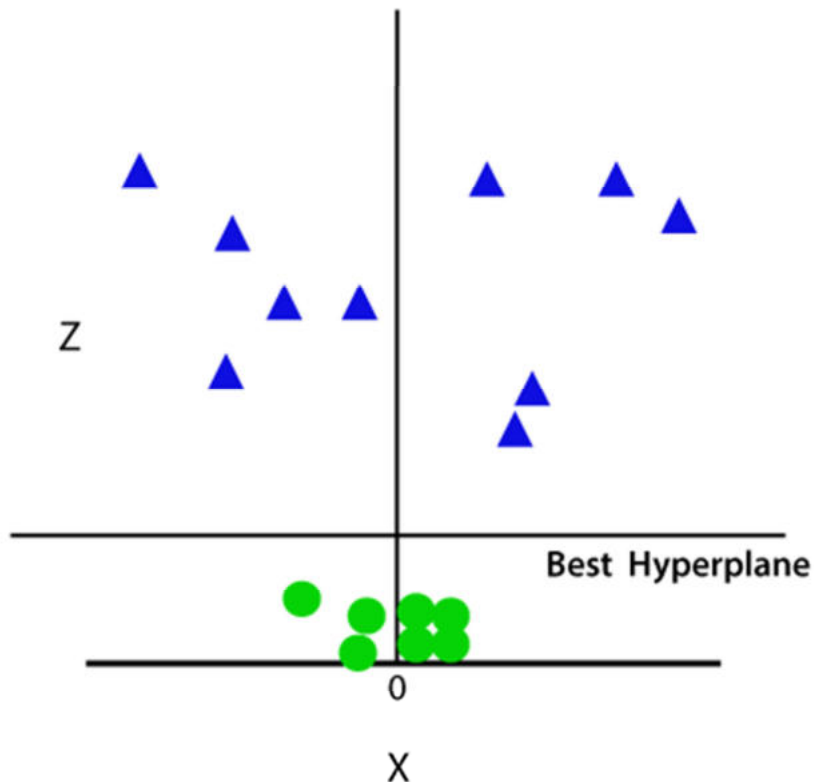
So to separate these data points, we need to add one more dimension. For linear data, we have used two dimensions x and y , so for non-linear data, we will add a third dimension z . It can be calculated as:

$$z = x^2 + y^2$$

By adding the third dimension, the sample space will become as below image:



So now, SVM will divide the datasets into classes in the following way. Consider the below image:

**Question:**

What does "learning from examples" mean in machine learning?

Ans: "Learning from examples" in machine learning refers to the process where a model improves its performance by analyzing and identifying patterns in given data samples. Instead of being explicitly programmed with rules, the model learns from input-output pairs (examples) to make predictions or decisions.

For example:

- In supervised learning, the model is trained using labeled data (e.g., images of cats and dogs with correct labels) to learn how to classify new, unseen images.
- In unsupervised learning, the model finds patterns in data without labeled examples (e.g., grouping customers based on purchasing behavior).

By iterating over numerous examples, the model generalizes from past observations to make accurate predictions on new data.

comparison table between **SVM** and **Logistic Regression** to help clarify when to use each:

Feature	Support Vector Machine (SVM)	Logistic Regression
Best for Data Type	Works well for both linear and non-linear data (using kernels)	Best for linearly separable data
Dimensionality	Performs well in high-dimensional spaces	Works well in low to moderate-dimensional data
Decision Boundary	Maximizes the margin (distance between classes)	Based on probability estimates using a sigmoid function

Handling Outliers	More robust to outliers due to margin maximization	More sensitive to outliers
Computational Complexity	Computationally expensive, especially with large datasets	Computationally efficient, especially for large datasets
Kernel Trick	Uses kernels (RBF, polynomial, etc.) for non-linear classification	Does not support kernel functions
Interpretability	Less interpretable, as it doesn't provide probability estimates	More interpretable, providing direct probability scores
Training Time	Slower, especially for large datasets	Faster and more scalable for large datasets
Use Case Examples	Text classification, image recognition, bioinformatics	Credit scoring, spam detection, medical diagnosis

When to Use Which?

- **Use SVM when:**
 - ✓ Data is high-dimensional
 - ✓ Non-linear relationships exist
 - ✓ Robustness to outliers is needed
 - ✓ You have a small-to-medium dataset
- **Use Logistic Regression when:**
 - ✓ Data is linearly separable
 - ✓ Probability predictions are required
 - ✓ Dataset is large and efficiency is important
 - ✓ You need an interpretable model

KNN classification algorithm

The K-Nearest Neighbors (KNN) classification algorithm is a straightforward, non-parametric method used in machine learning for pattern recognition. Here's a concise explanation of how it works:

Core Idea:

KNN classifies a new data point based on the majority class among its "k" nearest neighbors in the training dataset.

Steps:

1. **Choose K:**
 - Determine the value of "k," which represents the number of nearest neighbors to consider. This is a hyperparameter that must be set before running the algorithm.
2. **Calculate Distances:**
 - For a new, unlabeled data point, calculate the distance between it and all other data points in the training set. Common distance metrics include:

- Euclidean distance (straight-line distance)
- Manhattan distance (city block distance)
- 3. **Find Nearest Neighbors:**
 - Identify the "k" data points from the training set that have the smallest distances to the new data point. These are the "k" nearest neighbors.
- 4. **Determine Majority Class:**
 - Count the number of data points belonging to each class among the "k" nearest neighbors.
 - Assign the new data point to the class that appears most frequently (the majority class).
- 5. **Output Classification:**
 - The algorithm outputs the predicted class for the new data point.

Key Points:

- **Non-parametric:** KNN makes no assumptions about the underlying data distribution.
- **Lazy Learning:** It doesn't learn a model explicitly. It stores the training data and uses it at prediction time.
- **Distance-based:** The algorithm relies on distance calculations to find the nearest neighbors.
- **Sensitivity to K:** The choice of "k" significantly impacts the algorithm's performance.
- **Computational Cost:** Can be computationally expensive for large datasets, as it requires calculating distances to all training points for each prediction.
- **Feature Scaling:** It is very important to scale your features before running KNN, as features with larger ranges will dominate the distance calculations.

Example Scenario:

Imagine you have a dataset of flowers with features like petal length and petal width, and you want to classify a new flower as either a "Rose" or a "Tulip."

1. You choose $k = 5$.
2. You calculate the distances between the new flower and all the flowers in your dataset.
3. You find the 5 flowers that are closest to the new flower.
4. If 3 of those 5 are "Roses" and 2 are "Tulips," the algorithm classifies the new flower as a "Rose."

KNN Classifier Basics

- **Classification Focus:**
 - KNN is used to categorize data points into predefined classes.
 - It predicts the class of a new data point based on the classes of its "k" nearest neighbors in the training data.
- **"Lazy Learning":**
 - KNN is considered a "lazy learner" because it doesn't explicitly build a model during the training phase. Instead, it stores the training data and uses it at prediction time.

- **Distance Matters:**

- The algorithm relies on measuring the distance between data points. Common distance metrics include:
 - Euclidean distance
 - Manhattan distance

Model Example with a Simplified Dataset

Let's imagine a dataset of fruits, where we want to classify fruits as either "Apple" or "Orange" based on their "Sweetness" and "Size."

Dataset:

Fruit	Sweetness	Size	Class
Fruit 1	8	7	Apple
Fruit 2	7	6	Apple
Fruit 3	3	4	Orange
Fruit 4	2	3	Orange
Fruit 5	5	5	Apple
New Fruit	4	6	?

KNN Classification Process:

1. **Choose K:**
 - Let's set $K = 3$. This means we'll consider the 3 nearest neighbors.
2. **Calculate Distances:**
 - We need to calculate the distance between the "New Fruit" (Sweetness = 4, Size = 6) and all other fruits in the dataset. We'll use Euclidean distance:
 - $\text{Distance} = \sqrt{(\text{Sweetness}_2 - \text{Sweetness}_1)^2 + (\text{Size}_2 - \text{Size}_1)^2}$
 - After calculating the distances, we'll have:
 - $\text{Distance (New Fruit, Fruit 1)} = \sqrt{(8-4)^2 + (7-6)^2} = \sqrt{16 + 1} = \sqrt{17} \approx 4.12$
 - And so on for all of the other fruits.
3. **Find Nearest Neighbors:**
 - We then sort the distances and find the 3 nearest neighbors.
4. **Make a Prediction:**
 - Let's say the 3 nearest neighbors are:
 - Fruit 2 (Apple)
 - Fruit 5 (Apple)
 - Fruit 3 (Orange)
 - Because there are 2 apples and 1 orange, the KNN classifier predicts that the "New Fruit" is an "Apple."

Important Notes:

- **Choosing the right K value is crucial.** If K is too small, the model can be sensitive to noise. If K is too large, it can smooth out the decision boundaries.

- **Data preprocessing is important.** Features should be scaled to prevent features with larger ranges from dominating the distance calculations.
- **KNN can be computationally expensive for large datasets.**

Advantages of KNN:

- **Simplicity:**
 - KNN is easy to understand and implement. It doesn't involve complex mathematical formulas or extensive training processes.
- **Versatility:**
 - It can be used for both classification and regression tasks.
- **Non-parametric:**
 - KNN is a non-parametric algorithm, meaning it doesn't make assumptions about the underlying data distribution. This makes it suitable for data with complex, non-linear relationships.
- **No Training Period:**
 - It's a "lazy learner," so there's no explicit training phase. The model stores the training data and uses it directly for predictions.
- **Effective with Large Training Datasets:**
 - If you have a very large training dataset, the KNN algorithm can be very effective.

Disadvantages of KNN:

- **Computational Cost:**
 - Calculating the distance between the query point and all training data points can be computationally expensive, especially for large datasets. This makes it slow for real-time applications.
- **Memory Usage:**
 - KNN requires storing the entire training dataset in memory, which can be a problem for large datasets.
- **Sensitivity to Feature Scaling:**
 - The distance metric used by KNN is sensitive to the scale of the features. Features with larger ranges can dominate the distance calculations, leading to biased results. Therefore, feature scaling is crucial.
- **Choosing the Optimal K:**
 - Selecting the optimal value of K is crucial but can be challenging. A small K can lead to overfitting, while a large K can lead to underfitting.
- **Curse of Dimensionality:**
 - KNN's performance can degrade in high-dimensional spaces. As the number of features increases, the distance between data points becomes less meaningful, and the algorithm's accuracy decreases.
- **Imbalanced Data:**
 - In the case of imbalanced data, where one class dominates the others, KNN can be biased towards the majority class.

Question Bank:

1. What does "learning from examples" mean in machine learning?
2. Explain the three primary forms of learning: supervised, unsupervised, and reinforcement learning. Provide examples.
3. How does supervised learning differ from unsupervised learning?
4. What are the main applications of reinforcement learning?
5. Explain the concept of linear regression and its use in machine learning.
6. How does logistic regression differ from linear regression?
7. What is a Support Vector Machine (SVM)? How does it work?
8. In what scenarios would you use SVM over logistic regression?
9. What are some common applications of supervised and unsupervised learning?
10. Explain principle of Least Square? Explain metrics used for Linear regression like MAE, MSE, RMSE, R square.